OSKAR ORTIZ

Software Engineer | Backend | Java · Spring Boot · FastAPI · Python | AWS Cloud · DevOps

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PROFESSIONAL PROFILE

Software Engineering student in the sixth semester with hands-on academic and degree-project experience in backend development, REST APIs, relational databases, Docker and distributed AWS architectures. Technical background in Java, Spring Boot, Python, FastAPI, data engineering, cloud infrastructure, DevOps automation and applied AI.

TECHNICAL SKILLS

- Languages: Java, Python, SQL, object-oriented programming, exception handling and design patterns.
- Backend: Spring Boot, FastAPI, REST APIs, JPA/Hibernate, DTOs, MapStruct, validation and layered architecture.
- Databases: PostgreSQL, pgvector, MySQL, Amazon RDS and SQLite.
- Cloud and infrastructure: AWS VPC, public and private subnets, routing, Internet Gateway, NAT Gateway, EC2, Application Load Balancer, Auto Scaling, Lambda, RDS, SES, IAM, Security Groups and CloudWatch.
- DevOps and automation: Docker, Dockerfile, multi-stage builds, volumes, networks, Maven, Nginx, PowerShell, AWS CLI, Git and Git Flow.
- Data and AI: Pandas, PySpark, Selenium, ETL, exploratory data analysis, Power BI, Power Query, DAX, Random Forest, RAG, pgvector, Ollama and applied AI assistants.
- Frontend and tools: React, Next.js, Ionic, Jira, Azure DevOps, SonarQube, Figma, Balsamiq and UML.
- Networking and systems: Cisco Packet Tracer, routers, switches, LAN/WAN, IP addressing, Linux kernel fundamentals, processes, memory, I/O and firewalls.

SELECTED PROJECTS

CuyiElite - Degree Project

Technologies: *FastAPI, PostgreSQL, pgvector, Ionic, React, Random Forest, RAG*

- Designed, as Software Engineer and Tech Lead / Architect, a decision-support system aimed at reducing mortality in guinea pig production; project currently in development.
- Modeled a layered, offline-first architecture with local storage and synchronization, Random Forest classification and semantic search with pgvector.
- Planned a RAG engine over SENA manuals and field validation with producers at Centro Agropecuario Lope del SENA in Pasto.

University Enrollment System on AWS - Academic Project

Technologies: *AWS, VPC, EC2, ALB, Auto Scaling, RDS MySQL, Lambda, SES, IAM, CloudWatch, Python, Next.js, React, Nginx*

- Designed and deployed a distributed, scalable web application with layered architecture, public and private networks, traffic balancing and Auto Scaling.
- Configured high availability and database replication with RDS, plus an email microservice using Lambda and SES.
- Monitored the architecture with CloudWatch and configured AWS networking and security components.

Enrollment Deployment Automation - Academic Project

Technologies: *PowerShell, AWS CLI, JSON*

- Automated and validated deployment workflows with scripts for resource discovery, authentication checks and permissions review.
- Documented deployment evidence and AWS resource configuration.

Hotel Management Backend - Academic Project

Technologies: *Java, Spring Boot, Maven, PostgreSQL, JPA/Hibernate, REST, DTO, MapStruct*

- Implemented an object-oriented backend for customers, reservations, availability, prices, discounts, payments and notifications.
- Organized the solution with layered architecture, DTOs, MapStruct, validation and exception handling.
- Wrote unit tests for core application components.

Climate Data ETL Analysis - Academic Project

Technologies: *Python, Selenium, Pandas, SQLite, PySpark*

- Automated the download and validation of climate datasets and registered extraction activity.
- Cleaned and transformed data for SQLite storage and downstream processing.

Enterprise Network Simulation - Academic Project

Technologies: *Cisco Packet Tracer, routers, switches, LAN, WAN*

- Designed a topology connecting offices in different cities and configured routers, PCs and IP addressing.
- Validated connectivity between network segments using ping tests.

EDUCATION

Software Engineering - Universidad Cooperativa de Colombia, Pasto | Sixth semester in progress | Expected graduation: late 2027

CERTIFICATIONS

- Advanced Data Analysis Bootcamp - Talento Tech Región 3, 159 hours, August 2024.
- Machine Learning: Practical Examples in Sustainability and General Use - IBM SkillsBuild, July 2026. Credential ID: URL-5114BCC72FE5.
- Getting Started with Cisco Packet Tracer - Cisco Networking Academy, August 2026. Credential ID: 0f11de27-65ad-4f03-836d-495d3d12d633.

ENGINEERING FOUNDATIONS

- Design and architecture: encapsulation, inheritance, polymorphism, composition, Builder, Decorator, Adapter and Singleton patterns, SOLID, clean code and refactoring.
- APIs and systems: Controller-Service-Repository-DTO-Mapper-Domain architecture, REST endpoints, HTTP methods and input validation.
- Security: IAM, Security Groups and permission controls.

METHODOLOGIES AND LANGUAGES

Scrum | Kanban | Agile | Waterfall | Extreme Programming (XP)

- Technical English: reading and comprehension of technical documentation.